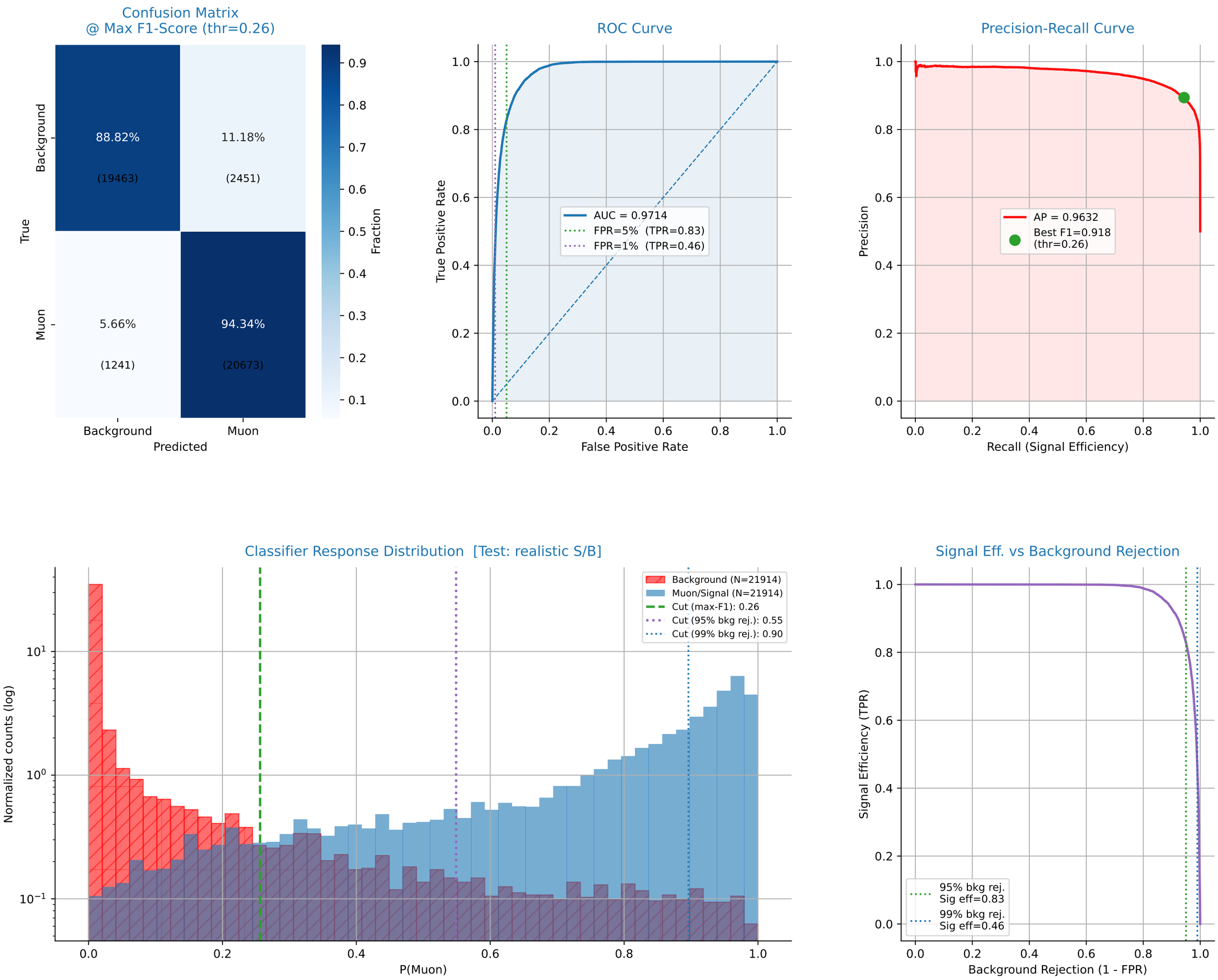
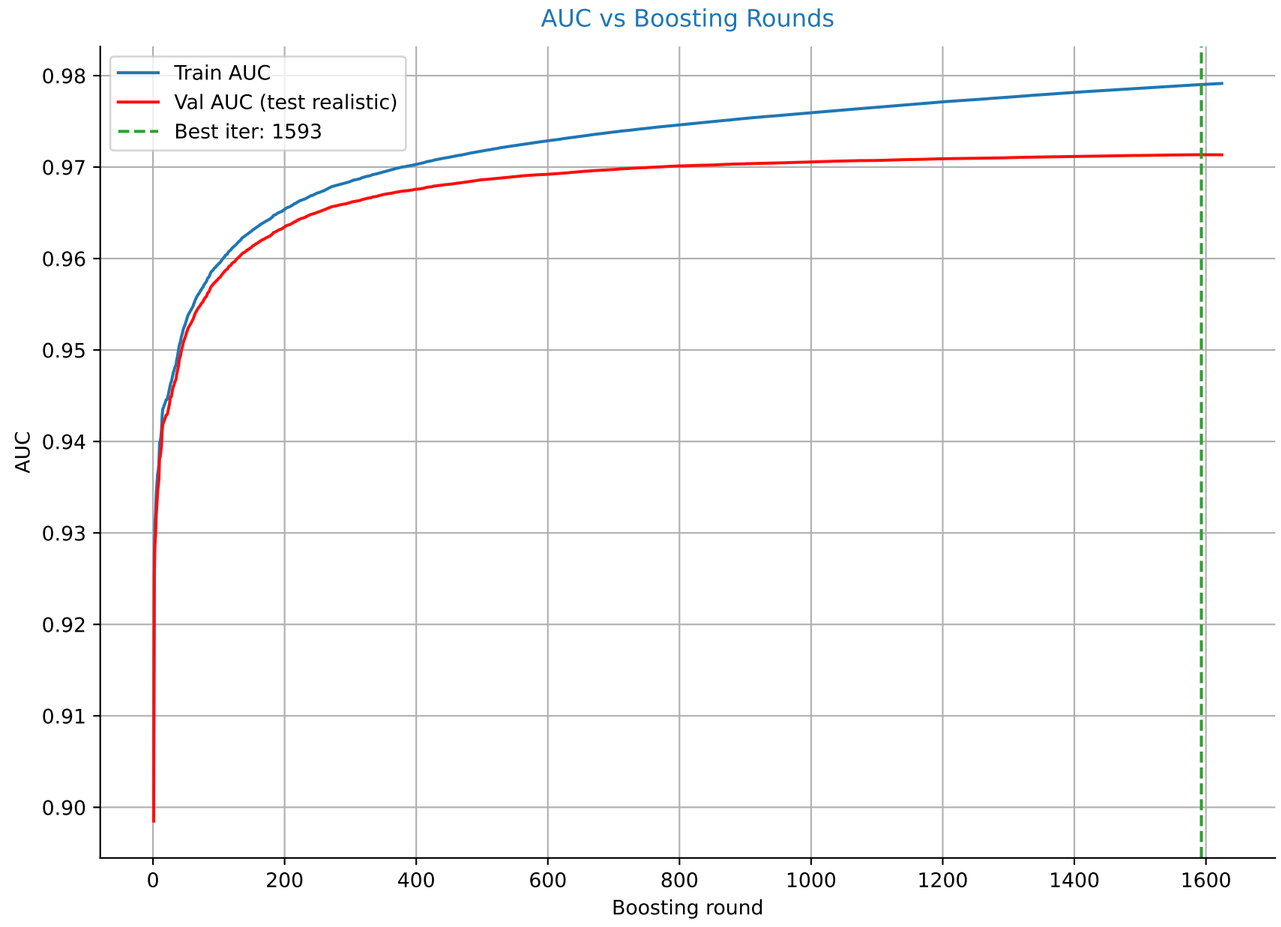
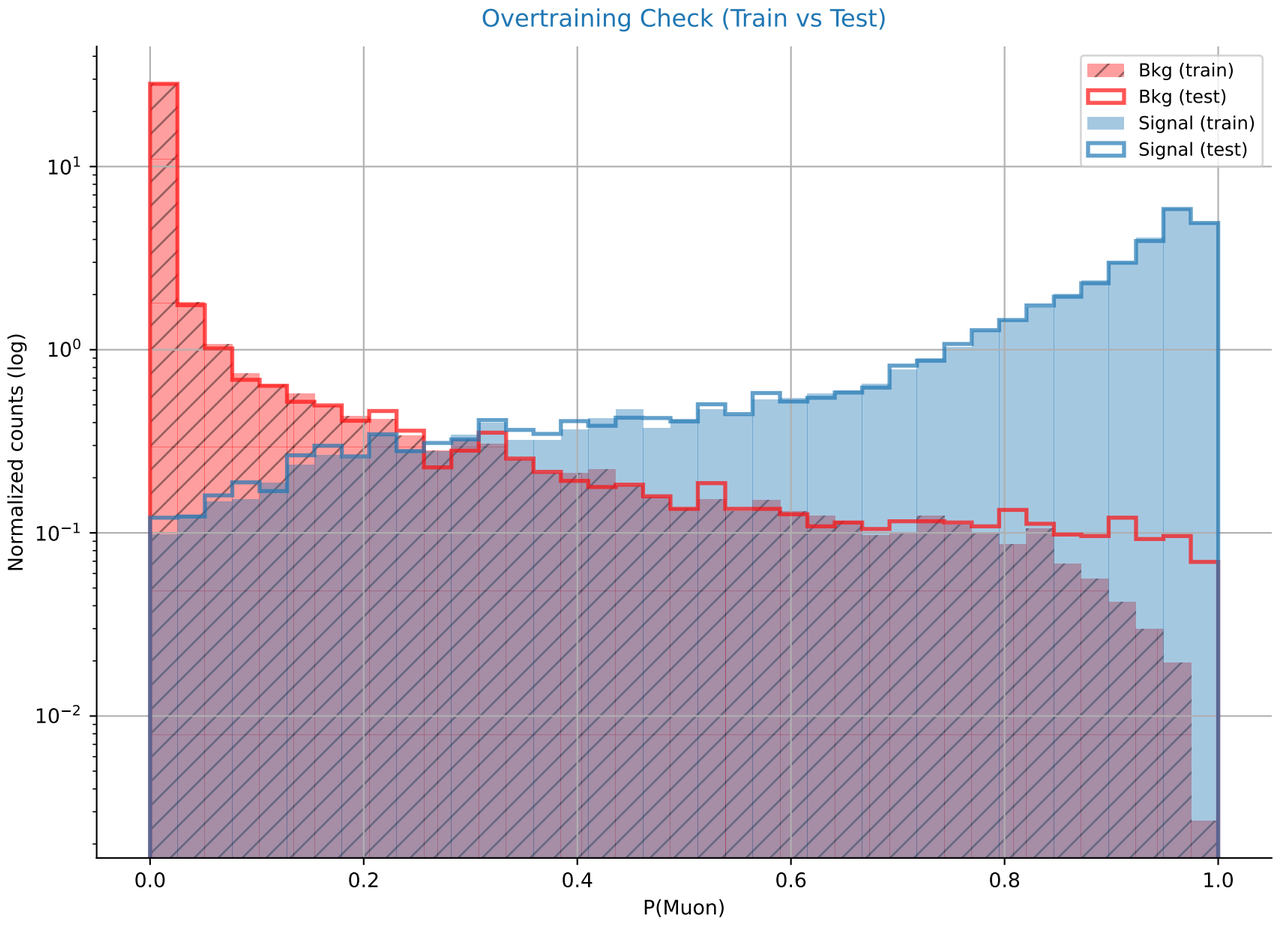
Muon Identification — XGBoost Analysis
[Train S/B=1:0 | Test S/B=1:1 (realistic)]

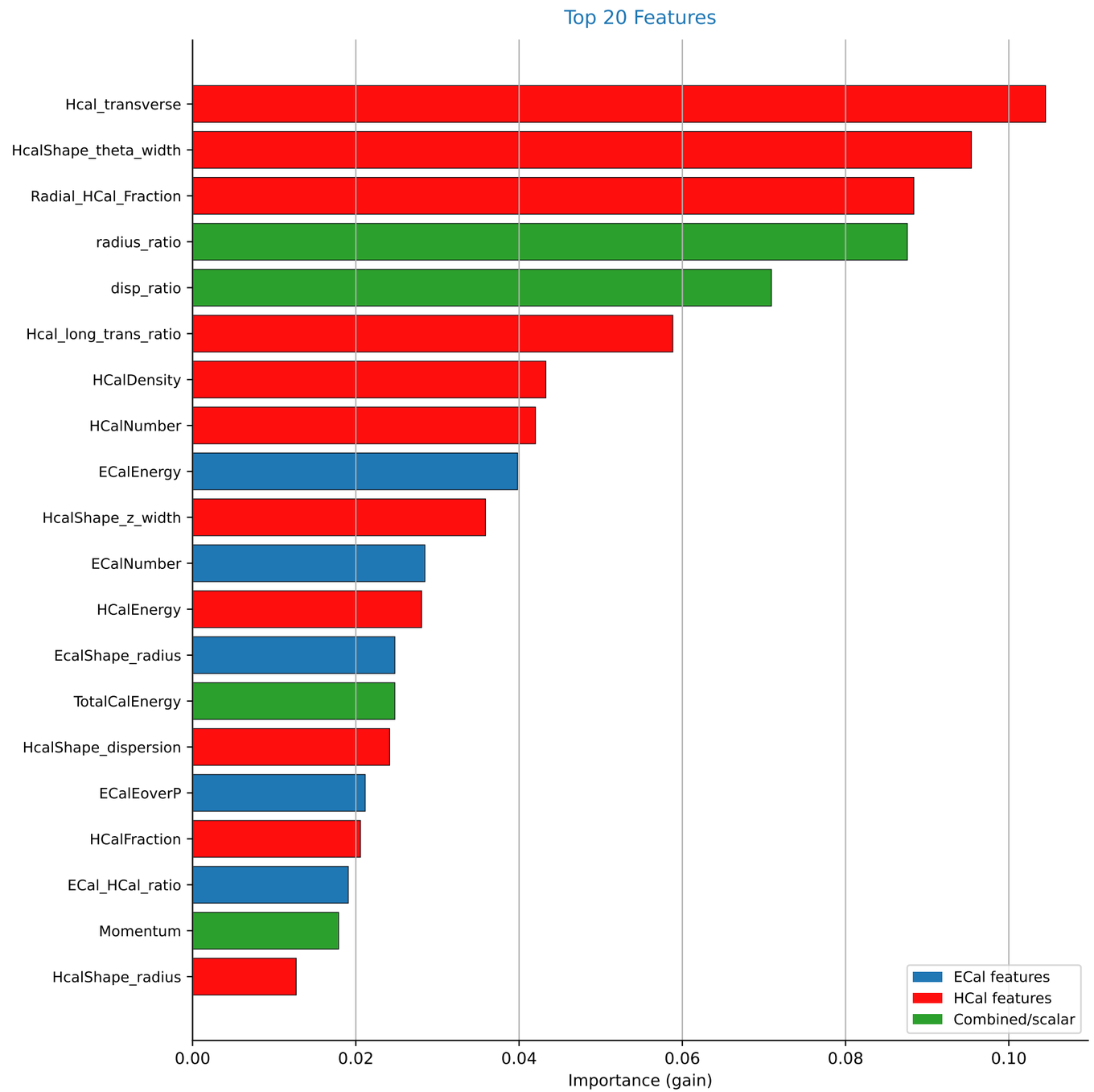


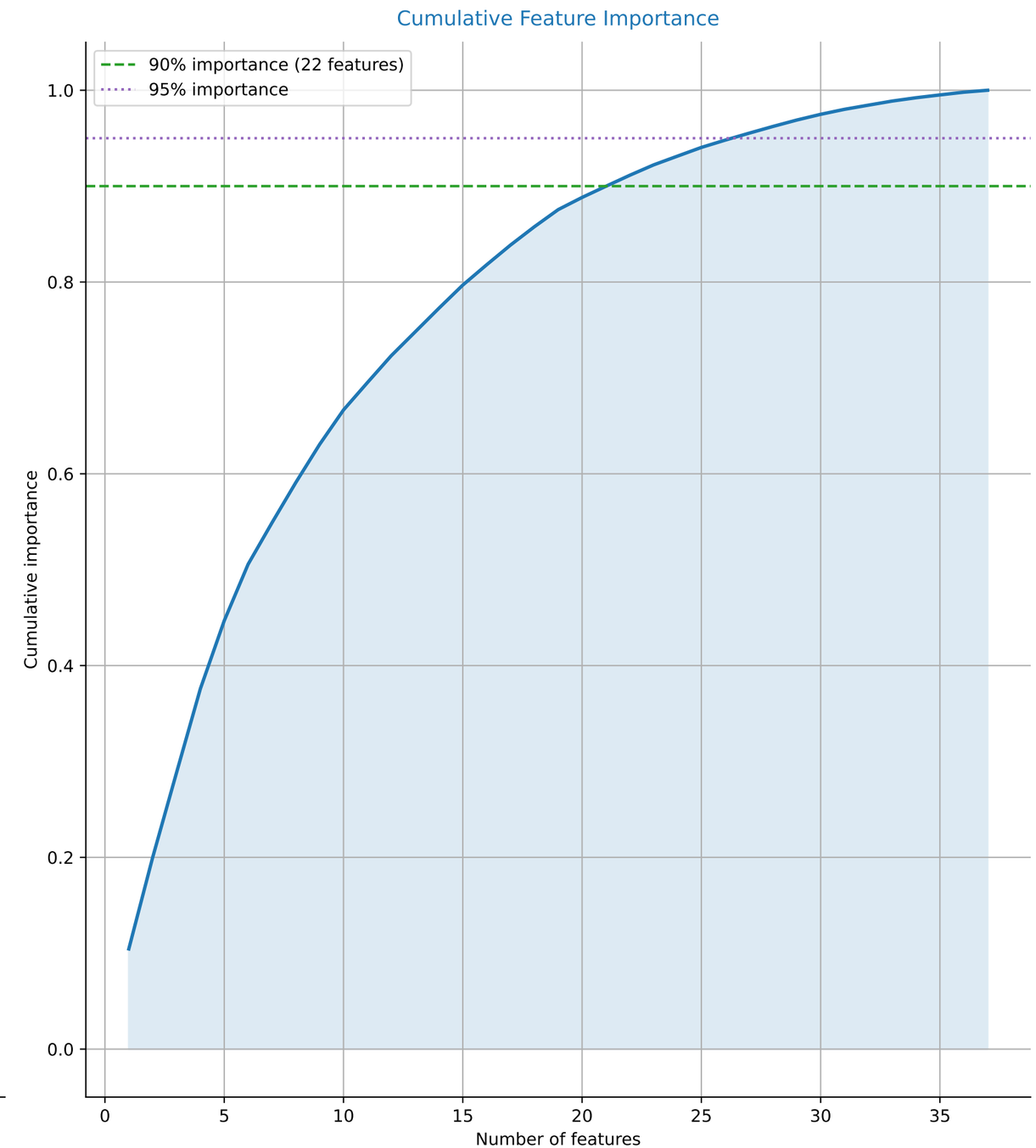
XGBoost Training Diagnostics



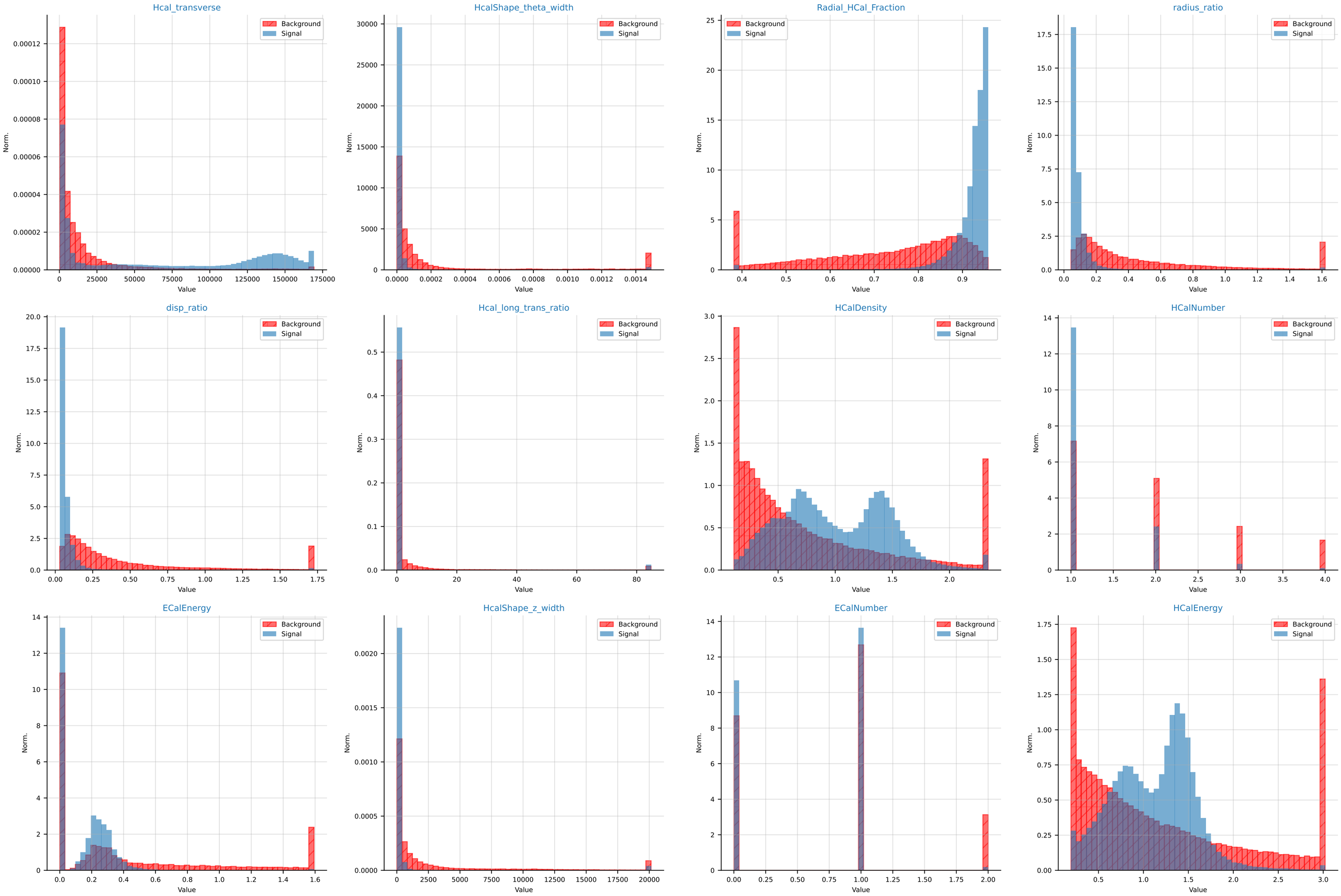


Feature Importances — XGBoost





Key Physics Variable Distributions: Signal vs Background
[rozkłady z całego datasetu przed podziałem]



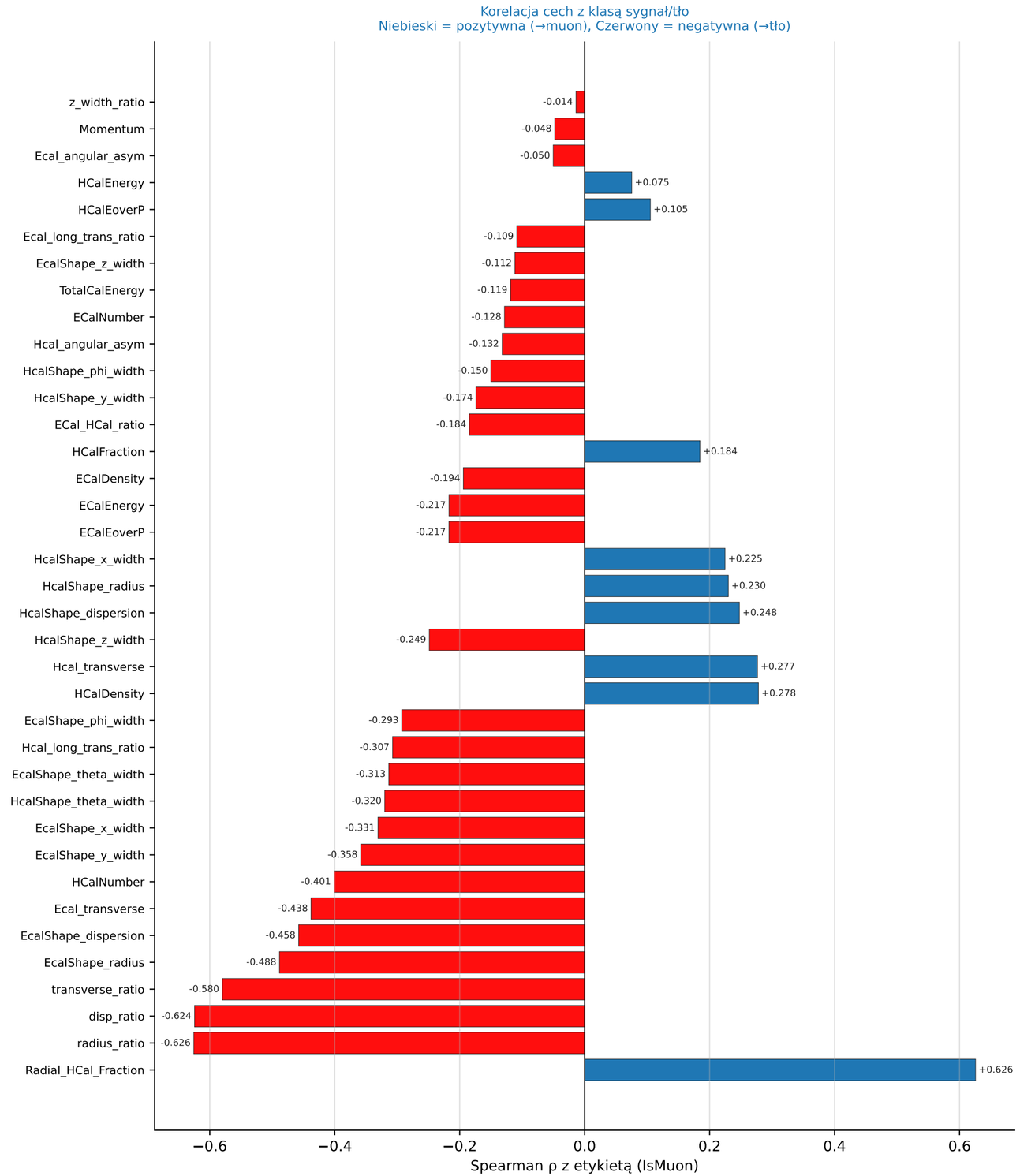
Spearman Correlation Matrix — All Features
(wartości z całego datasetu)

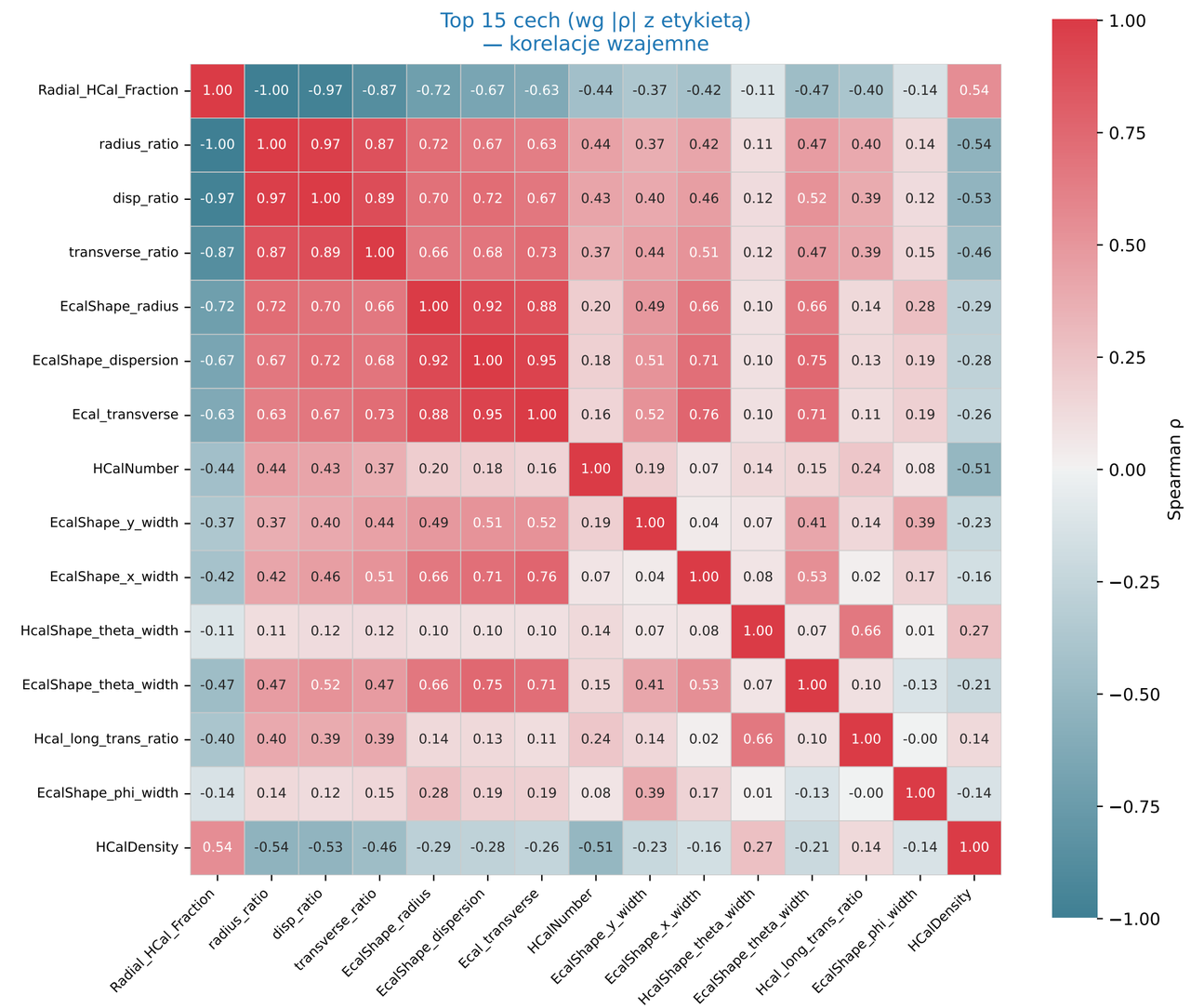
Wyższe |ρ| → silniejsza monotonna zależność między cechami

	ECalEnergy	HCalEnergy	ECalNumber	HCalNumber	ECalEoverP	HCalEoverP	Momentum	ECal_HCal_ratio	TotalCalEnergy	ECalDensity	HCalDensity	HCalFraction	radius_ratio	disp_ratio	z_width_ratio	Ecal_transverse	Hcal_transverse	transverse_ratio	Ecal_long_trans_ratio	Hcal_long_trans_ratio	Ecal_angular_asym	Hcal_angular_asym	Radial_HCal_Fraction	EcalShape_radius	EcalShape_dispersion	EcalShape_theta_width	EcalShape_phi_width	EcalShape_x_width	EcalShape_y_width	EcalShape_z_width	HcalShape_radius	HcalShape_dispersion	HcalShape_theta_width	HcalShape_phi_width	HcalShape_x_width	HcalShape_y_width	HcalShape_z_width
ECalEnergy	1.00																																				
HCalEnergy	0.33	1.00																																			
ECalNumber	0.90	0.41	1.00																																		
HCalNumber	0.13	0.05	0.10	1.00																																	
ECalEoverP	0.91	0.24	0.89	0.11	1.00																																
HCalEoverP	-0.18	0.35	-0.13	-0.05	0.04	1.00																															
Momentum	0.49	0.54	0.53	0.11	0.25	-0.50	1.00																														
ECal_HCal_ratio	0.95	0.17	0.90	0.10	0.93	-0.21	0.40	1.00																													
TotalCalEnergy	0.65	0.88	0.65	0.12	0.55	0.17	0.62	0.50	1.00																												
ECalDensity	0.99	0.34	0.88	0.12	0.90	-0.17	0.49	0.94	0.65	1.00																											
HCalDensity	0.20	0.79	0.29	-0.51	0.15	0.33	0.39	0.08	0.66	0.22	1.00																										
HCalFraction	-0.95	-0.17	-0.90	-0.10	-0.93	0.21	-0.40	-1.00	-0.50	-0.94	-0.08	1.00																									
radius_ratio	0.42	-0.28	0.26	0.44	0.50	0.00	-0.19	0.50	-0.01	0.38	-0.54	-0.50	1.00																								
disp_ratio	0.39	-0.28	0.26	0.43	0.50	0.04	-0.22	0.47	-0.03	0.35	-0.53	-0.47	0.97	1.00																							
z_width_ratio	-0.05	-0.01	-0.02	0.03	-0.05	-0.03	0.02	-0.05	-0.02	-0.06	-0.03	0.05	-0.03	-0.04	1.00																						
Ecal_transverse	0.45	-0.21	0.21	0.16	0.53	0.01	-0.23	0.43	0.04	0.43	-0.26	-0.43	0.63	0.67	-0.19	1.00																					
Hcal_transverse	0.44	0.53	0.57	-0.19	0.43	0.01	0.40	0.40	0.51	0.46	0.59	-0.40	-0.79	-0.78	-0.05	-0.23	1.00																				
transverse_ratio	0.37	-0.25	0.24	0.37	0.45	0.02	-0.18	0.43	-0.01	0.33	-0.46	-0.43	0.87	0.89	-0.14	0.73	-0.73	1.00																			
Ecal_long_trans_ratio	0.06	-0.11	0.03	0.08	0.08	-0.09	-0.04	0.09	-0.06	0.06	-0.13	-0.09	0.03	0.02	0.76	-0.04	-0.07	-0.11	1.00																		
Hcal_long_trans_ratio	0.46	0.34	0.52	0.24	0.49	0.07	0.26	0.44	0.43	0.46	0.14	-0.44	0.40	0.39	0.02	0.11	0.20	0.39	-0.02	1.00																	
Ecal_angular_asym	0.03	-0.06	-0.01	0.09	0.24	0.23	-0.29	0.05	-0.05	0.03	-0.10	-0.05	0.23	0.27	-0.05	0.35	-0.04	0.23	0.02	0.06	1.00																
Hcal_angular_asym	0.41	0.32	0.49	0.17	0.44	0.03	0.25	0.42	0.38	0.42	0.17	-0.42	0.10	0.13	-0.01	0.14	0.41	0.11	-0.02	0.55	0.31	1.00															
Radial_HCal_Fraction	-0.42	0.28	-0.26	-0.44	-0.50	-0.00	0.19	-0.50	0.01	-0.38	0.54	0.50	-1.00	-0.97	0.03	-0.63	0.79	-0.87	-0.03	-0.40	-0.23	-0.10	1.00														
EcalShape_radius	0.58	-0.21	0.24	0.20	0.56	-0.09	-0.15	0.53	0.11	0.56	-0.29	-0.53	0.72	0.70	-0.10	0.88	-0.29	0.66	0.05	0.14	0.28	0.09	-0.72	1.00													
EcalShape_dispersion	0.49	-0.21	0.22	0.18	0.56	0.01	-0.25	0.46	0.05	0.46	-0.28	-0.46	0.67	0.72	-0.10	0.95	-0.26	0.68	0.05	0.13	0.37	0.15	-0.67	0.92	1.00												
EcalShape_theta_width	0.34	-0.15	0.12	0.15	0.43	0.06	-0.24	0.32	0.03	0.33	-0.21	-0.32	0.47	0.52	-0.08	0.71	-0.15	0.47	0.06	0.10	0.76	0.20	-0.47	0.66	0.75	1.00											
EcalShape_phi_width	0.38	-0.12	0.16	0.08	0.09	-0.39	0.25	0.33	0.10	0.37	-0.14	-0.33	0.14	0.12	-0.03	0.19	-0.16	0.15	0.04	-0.00	-0.60	-0.31	-0.14	0.28	0.19	-0.13	1.00										
EcalShape_x_width	0.33	-0.16	0.15	0.07	0.39	-0.00	-0.18	0.32	0.02	0.31	-0.16	-0.32	0.42	0.46	-0.15	0.76	-0.12	0.51	-0.04	0.02	0.25	0.08	-0.42	0.66	0.71	0.53	0.17	1.00									
EcalShape_y_width	0.43	-0.14	0.18	0.19	0.37	-0.11	-0.06	0.37	0.10	0.41	-0.23	-0.37	0.37	0.40	-0.17	0.52	-0.23	0.44	-0.10	0.14	0.10	0.01	-0.37	0.49	0.51	0.41	0.39	0.04	1.00								
EcalShape_z_width	0.06	-0.12	0.03	0.09	0.07	-0.12	-0.03	0.10	-0.07	0.06	-0.14	-0.10	-0.00	-0.01	0.76	-0.06	-0.07	-0.14	0.96	-0.04	0.00	-0.05	0.00	0.03	0.03	0.04	0.07	-0.05	-0.08	1.00							
HcalShape_radius	0.51	0.58	0.65	-0.15	0.48	-0.02	0.47	0.46	0.57	0.52	0.60	-0.46	-0.85	-0.81	-0.04	-0.28	0.93	-0.72	-0.07	0.36	-0.08	0.52	0.85	-0.33	-0.31	-0.20	-0.13	-0.17	-0.24	-0.06	1.00						
HcalShape_dispersion	0.50	0.57	0.63	-0.15	0.47	-0.01	0.45	0.45	0.56	0.51	0.60	-0.45	-0.82	-0.81	-0.04	-0.25	0.95	-0.72	-0.07	0.35	-0.04	0.52	0.82	-0.31	-0.28	-0.17	-0.16	-0.15	-0.23	-0.07	0.99	1.00					
HcalShape_theta_width	0.45	0.40	0.50	0.14	0.48	0.12	0.24	0.40	0.48	0.44	0.27	-0.40	0.11	0.12	-0.00	0.10	0.41	0.12	-0.03	0.66	0.03	0.81	-0.11	0.10	0.10	0.07	0.01	0.08	0.07	-0.05	0.52	0.51	1.00				
HcalShape_phi_width	0.08	0.16	0.08	-0.08	0.06	0.08	0.11	0.02	0.16	0.08	0.21	-0.02	-0.06	-0.09	0.01	-0.14	0.15	-0.08	-0.03	0.14	-0.33	-0.38	0.06	-0.09	-0.15	-0.21	0.33	-0.09	0.00	-0.01	0.12	0.12	0.07	1.00			
HcalShape_x_width	0.26	0.37	0.35	-0.24	0.29	0.10	0.18	0.21	0.34	0.27	0.49	-0.21	-0.58	-0.57	-0.07	-0.07	0.72	-0.52	-0.09	0.08	0.02	0.12	0.58	-0.14	-0.09	-0.04	-0.16	0.03	-0.17	-0.09	0.64	0.67	0.18	0.21	1.00		
HcalShape_y_width	0.41	0.36	0.48	0.15	0.39	-0.01	0.33	0.39	0.42	0.41	0.22	-0.39	0.06	0.05	0.02	-0.08	0.42	0.00	-0.00	0.38	-0.05	0.54	-0.06	-0.04	-0.07	-0.05	0.05	-0.12	0.03	-0.00	0.46	0.45	0.60	0.11	-0.15	1.00	
HcalShape_z_width	0.46	0.43	0.53	0.13	0.48	0.07	0.31	0.42	0.49	0.46	0.29	-0.42	0.10	0.10	0.02	0.04	0.33	0.15	-0.04	0.95	0.01	0.55	-0.10	0.05	0.05	0.04	-0.00	0.01	0.07	-0.05	0.50	0.49	0.69	0.17	0.20	0.36	1.00

Spearman ρ: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Feature–Label Correlations & Clustered Heatmap





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║           Physics Analysis — Muon Identification Report           ║
║                        [ XGBoost ]                                ║
╚═══════════════════════════════════════════════════════════════════╝

  Input file      : MLDataGrape.root
  Total events    : 431055
  Signal (muons)  : 321481  (74.6%)
  Background      : 109574  (25.4%)
  Features used   : 37

────────────────────────────────────────────────────────────────────

  STRATEGIA PODZIAŁU:
    Trening  : 80% sygnału + 80% tła   →  S/B = 1:0.3
               Sygnał=257185  Tło=87660
    Test     : 10% puli syg. + 100% puli tła  →  S/B = 1:1.0
               Sygnał=21914   Tło=21914

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  MODEL: XGBoost
    n_estimators      = 3000  (best iter: 1593)
    learning_rate     = 0.05     max_depth         = 4
    min_child_weight  = 5        subsample         = 0.8
    colsample_bytree  = 0.8      gamma             = 1.0
    reg_alpha         = 0.1      reg_lambda        = 1.0
    scale_pos_weight  = 0.34

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  ROC AUC             : 0.9714
  Average Precision   : 0.9632

  Cut @ max-F1        : 0.256   (F1 = 0.918)
  Cut @ 95% bkg rej.  : 0.549   →  signal eff. = 0.828
  Cut @ 99% bkg rej.  : 0.896   →  signal eff. = 0.457

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  Classification report @ 95% bkg rejection threshold:

              precision    recall  f1-score   support

  Background       0.94      0.89      0.91     21914
        Muon       0.89      0.94      0.92     21914

    accuracy                           0.92     43828
   macro avg       0.92      0.92      0.92     43828
weighted avg       0.92      0.92      0.92     43828
```

Pełna numeryczna tabela ważności cech (XGBoost gain)
Wszystkie 37 cech posortowane malejąco | ρ(label) = korelacja Spearmana z etykietą (IsMuon)

Ranki 1 – 19

#	Feature	XGB Imp.	Cumul.	ρ(label)	Group
1	Hcal_transverse	0.10450	0.105	+0.277	HCal
2	Hcal_Shape_theta_width	0.09541	0.200	-0.320	HCal
3	Radial_HCal_Fraction	0.08837	0.288	+0.626	HCal
4	radius_ratio	0.08756	0.376	-0.626	Scalar/Combined
5	disp_ratio	0.07092	0.447	-0.624	Scalar/Combined
6	Hcal_long_trans_ratio	0.05884	0.506	-0.307	HCal
7	HCal_Density	0.04328	0.549	+0.278	HCal
8	HCal_Number	0.04201	0.591	-0.401	HCal
9	ECal_Energy	0.03982	0.631	-0.217	ECal
10	Hcal_Shape_z_width	0.03589	0.667	-0.249	HCal
11	ECal_Number	0.02847	0.695	-0.128	ECal
12	HCal_Energy	0.02806	0.723	+0.075	HCal
13	Ecal_Shape_radius	0.02478	0.748	-0.488	ECal
14	Total_HCalEnergy	0.02478	0.773	-0.159	Scalar/Combined
15	Hcal_Shape_dispersion	0.02415	0.797	+0.248	HCal
16	ECal_EoverP	0.02115	0.818	-0.217	ECal
17	HCal_Fraction	0.02057	0.839	+0.184	HCal
18	ECal_HCal_ratio	0.01908	0.858	-0.184	ECal
19	Momentum	0.01789	0.876	-0.058	Scalar/Combined

Ranki 20 – 37

#	Feature	XGB Imp.	Cumul.	ρ(label)	Group
20	Hcal_Shape_radius	0.01270	0.888	+0.230	HCal
21	Hcal_Shape_phi_width	0.01166	0.900	-0.150	HCal
22	Ecal_Shape_x_width	0.01148	0.911	-0.331	ECal
23	Hcal_Shape_x_width	0.01082	0.922	+0.225	HCal
24	Hcal_angular_asym	0.00913	0.931	-0.132	HCal
25	HCal_EoverP	0.00907	0.940	+0.105	HCal
26	Ecal_Shape_phi_width	0.00761	0.948	-0.293	ECal
27	Hcal_Shape_y_width	0.00723	0.955	-0.174	HCal
28	Ecal_transverse	0.00702	0.962	-0.438	ECal
29	Ecal_Shape_dispersion	0.00669	0.969	-0.458	ECal
30	ECal_Density	0.00596	0.975	-0.194	ECal
31	Ecal_angular_asym	0.00522	0.980	-0.050	ECal
32	Ecal_Shape_theta_width	0.00434	0.984	-0.313	ECal
33	Ecal_Shape_y_width	0.00427	0.989	-0.358	ECal
34	transverse_ratio	0.00346	0.992	-0.580	Scalar/Combined
35	Ecal_Shape_z_width	0.00294	0.995	-0.112	ECal
36	Ecal_long_trans_ratio	0.00285	0.998	-0.109	ECal
37	z_width_ratio	0.00202	1.000	-0.054	Scalar/Combined

ECal features | HCal features | Scalar / Combined